

Egor Sennov

Moscow, Russia

☎ +7 (967)-174-69-30 | ✉ esennov2005@gmail.com | 📷 SennovE | 🌐 sennov

Education

BS in Applied Mathematics and Computer Science

Moscow, Russia

NATIONAL RESEARCH UNIVERSITY – HIGHER SCHOOL OF ECONOMICS

September 2023 - June 2027

- **Relevant CS courses:** Python, Algorithms and Data Structures, Kotlin Mobile Development, C++, Distributed Systems, DBMS Development, Cloud Security
- **Mathematical courses:** Linear Algebra and Geometry, Calculus, Discrete Mathematics, Number Theory, Machine Learning

Experience

T-Bank: Media Projects Team

Moscow, Russia

BACKEND ENGINEER

March 2025 - Present

The team responsible for news and user-generated content services at T-Bank. Built and maintained content APIs and internal tools.
Stack: Python, Postgres, Redis.

- Integrated the airline ticket service from T-Travel into the bank's magazine, allowing for recommendations on current tickets directly in articles.
- Rewrote the server using Nest JS and refactored it to handle market data within the bank's journal. Conducted a seamless transition from the old server to the new one.
- Refactored the survey service, which allowed to remove unused data from the database, significantly reduce and optimize the service's code, and greatly simplify the process of adding new surveys.
- Also assisted the T-Investments team with the development of a Python SDK for trading. Wrote an auto-generator for classes from proto files for sending grpc requests.

Yandex: Plus and Fantech DWH Content Services Team

Moscow, Russia

DATA ENGINEER

October 2024 - February 2025

The team develops ETL processes covering the main data domains, and also helps to develop the architecture of product development services.
Stack: Python, SQL, YQL, ClickHouse, YTsaurus.

- Optimized the operation of data pools by reducing CPU consumption, which reduced the execution time of ETL chains.
- Accelerated data collection for analytics by switching tasks to using more efficient algorithms.
- Was the first member of the team to use ClickHouse functions, which improved the accuracy of calculations and significantly improved the speed of aggregations.
- Has built processes for working with new data sources, including processing new logs and offline downloads.
- Unloaded the backend capacity by transferring calculations to our team's side.

Yandex Backend Development School

Moscow, Russia

BACKEND DEVELOPER

Summer 2024

The team has been developing a platform for creating interactive longreads using FastAPI and PostgreSQL, which supports multimedia and user progress tracking.

- Implemented an authorization system based on FastAPI and Passlib, implemented JWT tokens for access control.
- Used SQLAlchemy to work with PostgreSQL via ORM, configured automatic migrations with Alembic.
- Integrated Yandex Object Storage for storing files using Aioboto3, made endpoints for secure file management.
- Configured fault tolerance using Nginx for load balancing.
- Wrote automated tests with Pytest to test the key features of the platform.

Skills

Programming Languages Python (3.13), C++ (23), SQL, C, Assembler x86-64, HTML, CSS, JavaScript, Kotlin, MQL4, Solidity
Software & Tools Git, LaTeX, ClickHouse

Projects

Pre-commit-hook for SQL code validation and a command-line interface for formatting and receiving AI-powered optimization recommendations. [↗](#)

- An AI-integrated command-line interface that optimizes SQL queries based on database analysis and resource usage. You can get a generated web page describing the improvements and indicating the query bottlenecks.
- A pre-commit hook that easily integrates into your project and automatically runs checkers and optimizations on every Git commit.

A robot advisor that simplifies trading in the terminal RoboForex - MetaTrader 4 [↗](#)

- A bot that allows you to automate the placing and closing of orders with customizable parameters
- An interface that can be connected to a trading terminal and used to control a bot

Telegram bot with an up-to-date schedule for BAMl students at the HSE

- A bot class has been developed to simplify working with the python-telegram-bot library
- Implemented automatic updating of the lesson schedule and registration of users in the database

A module for working with polynomials in python

- A class of polynomials has been created from scratch, which contains terms with different degrees of the variable and the remainder in the form of a regular irreducible fraction
- Implementations of FFT and IFFT algorithms, gcd finding functions for polynomials have been created
- The module also makes it possible to work with fractions

Program for editing BMP files

- The program is written in raw C++ without the use of third-party libraries for image processing.
- Filters have been created that process images by applying matrices to pixels, as well as using more complex algorithms to implement their functionality.
- Unit-tests have been written to verify the correctness of the main program and its filters.

Extracurricular ---

Языки

- Russian (native)
- English (fluent)
- German (elementary)